

JULIAN ZHANG

+1 (647) 914-2970 | julian.zhang@uwaterloo.ca | [linkedin.com/in/zhaju](https://www.linkedin.com/in/zhaju) | zhaju.github.io

EDUCATION

University of Waterloo

(Sep 2024 - Apr 2029)

Bachelor of Computer Science and Finance Double Major (CFM) | *President's Scholarship of Distinction*

University of Toronto Schools

(Sep 2018 - Jun 2024)

Ontario Secondary Schools Diploma | *Cumulative GPA: 4.0*

EXPERIENCE

Oakville Centre For Vision - Software Engineer

(May 2024 - Sep 2024)

- Created a web platform for eye-tracking goggles used to analyze vision patterns in professional athletes, collaborating directly with optometrist Dr. Ronald Gall
- Focused on data analysis of **8+ tests** that examine gaze vectors for saccades, fixations, and pursuits
- Collaborated with a hardware team to modernize a **6,000+ line** codebase for improved functionality

University of Toronto Clean Energy Lab - Research Assistant

(Jun 2022 - Sep 2023)

- Applied Crystal Graph Convolutional Neural Network (CGCNN) to examine trends in chemical properties
- Analyzed results for material discovery with a focus on advancing lithium-ion alternatives for sustainable energy. Synthesized findings for practical applications of mitigating climate change
- Presented findings to Prof. Oleksandr Voznyy and PhD Students as the group's youngest member

Canadian National Debate Team - Member

(May 2023 - Aug 2024)

- Represented Canada as **1 of 12** debaters at the 2024 Belgrade World School Debating Championship (WSDC) 2024, honing advanced critical analysis, structured thinking, and public speaking skills
- Conducted research across politics, economics, and international relations, engaging with field experts
- Organized 5+ debate tournaments attracting **500+** participants and raising **\$10,000+** for charity

PROJECTS

- SignWave (JamHacks Winner):** AI translator converting text and audio into sign language animations, improving accessibility for the deaf and hard of hearing (OpenCV, Flask, MediaPipe, Whisper)
- EdgeFund:** Digital stock broker using statistical methods to select a portfolio of stocks, outperforming the TSX60 and S&P500 by more than **30% over 1 year** (Yahoo Finance, Pandas, NumPy, matplotlib)
- zhaju.github.io:** Personal website featuring a blog and repository of high school and university course notes, supporting **100+ students** with educational content (HTML/CSS, TypeScript, Node.js, Quartz)

SKILLS

- Languages:** C/C++/C#, Python, Racket, Java, JavaScript, HTML/CSS, LaTeX
- Libraries:** React.js, Node.js, NumPy, matplotlib, Pandas, PyTorch, TensorFlow, Yahoo Finance
- Coursework:** Algorithm Design and Data Abstraction, Financial Markets and Data Analytics
- Certifications:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning

SELECTED AWARDS

- Canadian National British Parliamentary Debate Championship - **1st speaker & team**
- Munk Debates ReSolved Global Debate Competition - **Champion, awarded \$5,000**
- JamHacks VII Hackathon (300+ competitors) - **First Prize Overall, Best Use of GitHub**
- American Invitational Math Exam (AIME) Qualifier - **Top 8% Worldwide**